

・三角関数の不定積分～倍角の公式～

(例) 次の不定積分を求めよ。

$$(1) \int (\sin x + \cos x)^2 dx \quad (2) \int x \sin x \cos x dx \quad (3) \int \frac{\cos 2x}{\sin x - \cos x} dx$$

$$(4) \int \sin^2 x dx \quad (5) \int \cos^4 x dx \quad (6) \int x \sin^3 x dx$$

$$(7) \int \cos 2x \tan x dx$$

point

積分は積・商に弱い。和・差に強い。

$$\begin{aligned} & \text{2倍角} \begin{cases} \sin x \cos x = \frac{1}{2} \sin 2x \\ \sin^2 x = \frac{1}{2} (1 - \cos 2x) \\ \cos^2 x = \frac{1}{2} (1 + \cos 2x) \end{cases} \\ & \text{3倍角} \begin{cases} \sin^3 x = \frac{1}{4} (3 \sin x - \sin 3x) \\ \cos^3 x = \frac{1}{4} (3 \cos x + \cos 3x) \end{cases} \end{aligned}$$

Cは積分定数とする。

$$\begin{aligned} (1) \quad & \int (\sin x + \cos x)^2 dx \\ &= \int (\sin^2 x + 2 \sin x \cos x + \cos^2 x) dx \\ &= \int (1 + \sin 2x) dx \\ &= x - \frac{1}{2} \cos 2x + C_{11} \\ (2) \quad & \int x \sin x \cos x dx \\ &= \int x \cdot \frac{1}{2} \sin 2x dx \\ &= \frac{1}{2} \int x \cdot \left(-\frac{1}{2} \cos 2x\right)' dx \\ &= \frac{1}{2} \left(-\frac{1}{2} x \cos 2x + \frac{1}{2} \int 1 \cdot \cos 2x dx\right) \\ &= -\frac{1}{4} x \cos 2x + \frac{1}{8} \sin 2x + C_{11} \end{aligned}$$

$$\begin{aligned} (3) \quad & \int \frac{\cos 2x}{\sin x - \cos x} dx \\ &= \int \frac{\cos^2 x - \sin^2 x}{\sin x - \cos x} dx \\ &= \int \frac{(\cos x + \sin x)(\cos x - \sin x)}{\sin x - \cos x} dx \\ &= - \int (\cos x + \sin x) dx \\ &= -\sin x + \cos x + C_{11} \\ (4) \quad & \int \sin^2 x dx \\ &= \frac{1}{2} \int (1 - \cos 2x) dx \\ &= \frac{1}{2} \left(x - \frac{1}{2} \sin 2x\right) + C \\ &= \frac{1}{2} x - \frac{1}{4} \sin 2x + C_{11} \end{aligned}$$

$$\begin{aligned} (5) \quad & \int \cos^4 x dx \\ &= \int \left(\frac{1 + \cos 2x}{2}\right)^2 dx \\ &= \frac{1}{4} \int (1 + 2 \cos 2x + \cos^2 2x) dx \\ &= \frac{1}{4} \int \left(1 + 2 \cos 2x + \frac{1 + \cos 4x}{2}\right) dx \\ &= \frac{1}{4} \int \left(\frac{3}{2} + 2 \cos 2x + \frac{1}{2} \cos 4x\right) dx \\ &= \frac{1}{4} \left(\frac{3}{2} x + \sin 2x + \frac{1}{8} \sin 4x\right) + C \\ &= \frac{3}{8} x + \frac{1}{4} \sin 2x + \frac{1}{32} \sin 4x + C_{11} \\ (6) \quad & \int x \sin^2 x dx \\ &= \frac{1}{4} \int x (3 \sin x - \sin 3x) dx \\ &= \frac{1}{4} \int x \left(-3 \cos x + \frac{1}{3} \cos 3x\right)' dx \\ &= \frac{1}{4} \left\{ x \left(-3 \cos x + \frac{1}{3} \cos 3x\right) - \int 1 \cdot \left(-3 \cos x + \frac{1}{3} \cos 3x\right) dx \right\} \\ &= \frac{1}{4} \left(-3x \cos x + \frac{1}{3} x \cos 3x + 3 \sin x - \frac{1}{9} \sin 3x\right) + C \\ &= -\frac{3}{4} x \cos x + \frac{1}{12} x \cos 3x + \frac{3}{4} \sin x - \frac{1}{36} \sin 3x + C \end{aligned}$$

$$\begin{aligned} (7) \quad & \int \cos 2x \tan x dx = \int (2 \cos^2 x - 1) \frac{\sin x}{\cos x} dx \\ &= \int \left\{ \sin 2x + \frac{(\cos x)'}{\cos x} \right\} dx \\ &= -\frac{1}{2} \cos 2x + \log |\cos x| + C_{11} \end{aligned}$$